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Q1. DICTIONARY

ANSWER

```
cubes={n:n**3 for n in range (1,11)}
print(cubes)
OUTPUT
{1:1,2:8,3:27,4:64,5:125,6:216,7:343,8:512,9:729,10:1000}
```

Q2. DICTIONARY

ANSWER

```
nums=[1,2,3,4,5]
result={"even:" [], "odd:" []}
for n in nums:
    if n % 2 == 0:
        result["even"].append(n)
    else:
        result["odd"].append(n)
print(result)
OUTPUT
{"even": [2,4,6], "odd": [1,3,5]}
```

Q3. Tuple

ANSWER

```
LIST TO TUPLE
LST=[1,2,3]
t=tuple(LST)
print(t)
TUPLE TO LIST
T=[1,2,3]
LST=LIST(T)
print(LST)
WE USE TUPLE TO LIST WHEN WE NEED TO MODIFY THE DATA
WE USE LIST TO TUPLE WHEN WE WANT TO FIX/IMMUTABLE DATA
```

Q4. LIST

ANSWER

```
5  10  15  20  25  30
-6 -5   -4  -3  -2   -1
a)print([-3:])
b)print([: : -1])
c)print([1:-1])
OUTPUTS
a)20 25 30
b)30 25 20 15 10 5
c)10 15 20 25
```

Q5. DICTIONARY

ANSWER

```
searching in list
for example:
numbers=[10,20,30,40,50]
print(30 in numbers)
in this case python will check elements one by one like 10 after 20 after 30 it will check each and every element
searching in dictionary
for example:
student={"id1":"Ravi","id2':"Yash","id3":"Raj"}
print("id2" in student)
dictionary uses keys to find values and it uses hashing to locate the key quickly
list average time complexity is 0(n)
dictionary average time complexity is 0(1)
```